Executive Summary: Telecom Customer Churn Prediction

Objective

Identify the key factors driving customer churn at a telecom company and build a predictive model to flag customers at high risk of leaving, enabling proactive retention efforts.

Approach

Analyzed a dataset of 7,043 telecom customers covering demographics, account details, and subscribed services. Approximately 26.5% of customers in the dataset had churned. After exploratory analysis, five classification models — Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and Gradient Boosting — were trained and evaluated on a held-out test set to identify the most effective approach for predicting churn.

Key Findings

1. **Contract length is the single biggest churn driver.** Customers on month-to-month contracts churn at substantially higher rates than those on annual or two-year contracts, indicating that contract flexibility without incentive to commit long-term increases attrition risk.

2. **Early tenure is a high-risk window.** Customers who have been with the company for only 1–2 months churn far more often than long-tenured customers, suggesting the first few months of the customer relationship are critical for retention intervention.

3. **Service bundling reduces churn.** Customers without add-on services such as Online Security, Online Backup, Device Protection, or Tech Support churn more frequently than those who subscribe to these services, suggesting bundled services increase switching costs and customer stickiness.

4. **Payment method correlates with churn.** Customers paying via electronic check churn more than those using automatic payment methods, which may reflect either a less "locked-in" customer relationship or friction in the payment experience.

5. **Fiber optic internet customers churn more** than DSL customers, which may point to service quality, pricing, or competitive pressure specific to that segment.

Model Performance

Five models were compared on Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Gradient Boosting and Logistic Regression performed comparably overall (ROC-AUC ≈ 0.84), with Logistic Regression correctly identifying a higher share of actual churners (56.7% recall vs. 50.8% for Gradient Boosting). Given that failing to identify an at-risk customer typically carries a higher business cost than a false alarm, this recall trade-off is an important consideration in model selection — not just overall accuracy.

Business Recommendations

- **Incentivize contract commitment:** offer discounts or loyalty benefits to move month-to-month customers onto longer-term plans.
- **Strengthen early-tenure engagement:** build a structured onboarding and check-in process for customers in their first 1–2 months.
- **Promote service bundles:** proactively offer Online Security, Backup, Device Protection, and Tech Support to customers who don't currently have them.
- **Review the electronic check payment experience** and encourage migration to automatic payment methods.
- **Deploy the model operationally** to score the active customer base monthly and route high-risk customers to a targeted retention campaign.

Next Steps

To move this analysis from a baseline model to a production-ready solution, recommended next steps include addressing class imbalance to improve detection of churners, tuning model hyperparameters, validating performance with cross-validation, and incorporating explainability (SHAP) so retention teams can understand *why* a given customer is flagged as high-risk.